

PROJECT REPORT

Project Title: Detecting Data Leaks Using SQL Injection

Task number: 02

Objective:

The objective of this is to develop a secure user authentication system that protects sensitive information from unauthorized access and prevents data leaks. The system uses password encryption and double-layer verification to enhance security.

Problem Statement:

SQL injection attacks are one of the common threats to databases and web applications. Attackers may attempt to access confidential information by manipulating input fields. Therefore, a secure mechanism is required to protect user credentials and sensitive data.

Proposed Solution:

The proposed system provides:

- User registration facility.
- Password encryption before storage.
- Password decryption during authentication.
- Double-layer verification using username and password.
- Protection against unauthorized access and data leakage.

Technologies Used:

- Programming Language: Java
- IDE: Eclipse
- Concepts Used:
 - Encryption and Decryption
 - User Authentication
 - String Manipulation
 - Double-Layer Security
 -

Modules

1. AESUtil Class:

This class performs encryption and decryption of passwords.

Functions:

- encrypt(String password)
- decrypt(String password)

The encryption process shifts each character by +3, and decryption shifts it by -3.

2. RegisterUser Class:

This class stores the username and encrypted password.

Functions:

- register(String uname, String pass)

Responsibilities:

- Accept user credentials.
- Encrypt the password.
- Store encrypted data.

3. LoginUser Class:

This class authenticates users.

Responsibilities:

- Accept login credentials.
- Decrypt stored password.
- Verify username and password.
- Allow login if credentials are valid.

Working Process of the Project

Step 1: User Registration

- The user enters a username and password.
- Example:

- Username: Pallavi
- Password: abc

Step 2: Password Encryption

- The AESUtil.encrypt() method encrypts the password by shifting each character by **+3**.
- Example:
 - Original Password: abc
 - Encrypted Password: def

Step 3: Store User Details

- The RegisterUser class stores:
 - Username: Pallavi
 - Encrypted Password: def

Step 4: User Login

- The user enters the login credentials.
- Example:
 - Username: Pallavi
 - Password: abc

Step 5: Password Decryption

- The stored encrypted password (def) is decrypted using the AESUtil.decrypt() method.
- Each character is shifted by **-3**.
- Example:
 - Encrypted Password: def
 - Decrypted Password: abc

Step 6: Authentication

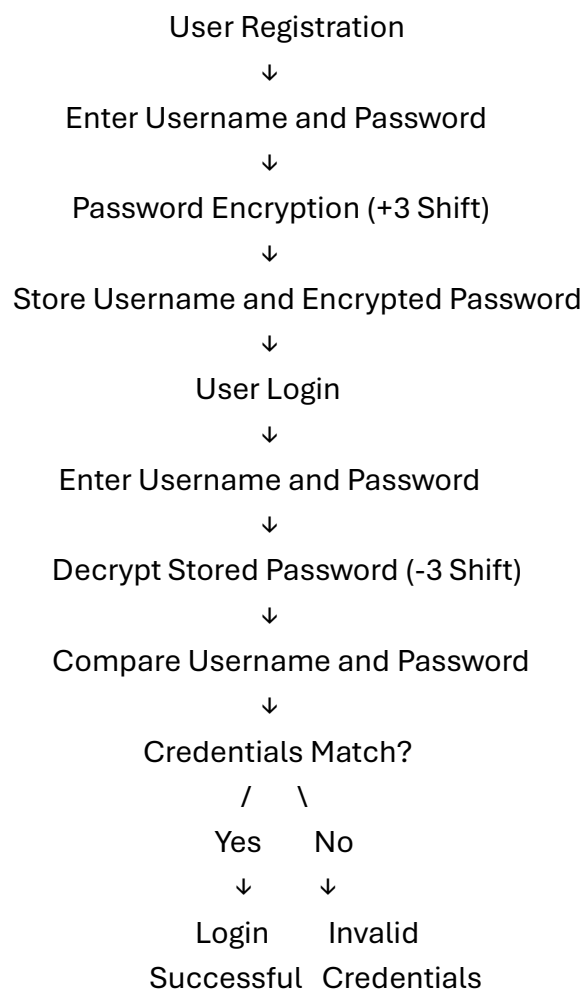
- The system compares:
 - Entered username with stored username.

- Entered password with decrypted password.

Step 7: Access Decision

- If both username and password match:
 - **Login Successful**
- Otherwise:
 - **Invalid Credentials**

Flow Diagram



Sample Execution

Input:

Username = Pallavi

Password = abc

Registration Output:

User Registered Successfully

Encrypted Password = def

Login Input:

Username = Pallavi

Password = abc

Login Output:

Login Successfu

Applications:

- Login Systems
- Banking Applications
- Educational Portals
- Employee Management Systems
- Secure User Authentication System

Conclusion:

This project demonstrates a secure login system using Java. Passwords are encrypted before storage and verified during login. The system provides double-layer security through username and password authentication and serves as a foundation for implementing advanced mechanisms to prevent SQL injection and data leaks.